Problems and Solutions

The set of information that defines the problem and any constraints on the solution is the *problem domain* (or sometimes, the *problem space*). The problem domain is a description of the real-world entities, processes, constraints, etc. that the software must model. These could include people (often with different roles to perform), commercial goods and services, resources (e.g., meeting rooms, physical documents), statutory time limits (perhaps the software manages legal casework, and certain documents have to be filed on time), and so on.

By considering the problem domain in terms of the entities in the system (together with their actions, their relationship to each other, and especially the outcomes that need to be achieved), the requirements for the solution – the software system we are creating – will emerge. When all the requirements are met, we have our solution, and the environment in which the solution exists is known as the *solution space*. The solution space is an abstraction that encompasses the process of developing the solution, the environment where it is built (the code we write, the server it runs on), and most importantly, its design.

The processes and techniques involved in analyzing businesses and arriving at a useful set of functional requirements for a software engineer to act upon are beyond the scope of this book (and there are many books out there on the topic), but we can certainly look at how to apply OOD to requirements.

Modeling

OOP software is organized into classes that represent real entities, the actions performed upon them, the services required for the software to function in its environment, and so on. From analyzing the problem domain, we learn which abstractions we will need. For example, suppose we had to create software to manage a chain of bookshops that also provided online retail services. Clearly, we will need to represent shops and books for starters. We will likely also need to represent customers, employees, warehouses, and so on.

Let's invent some formal requirements.

- We sell books.
- Multiple bricks-and-mortar shops with local inventory.
- Each shop has a manager and multiple employees.